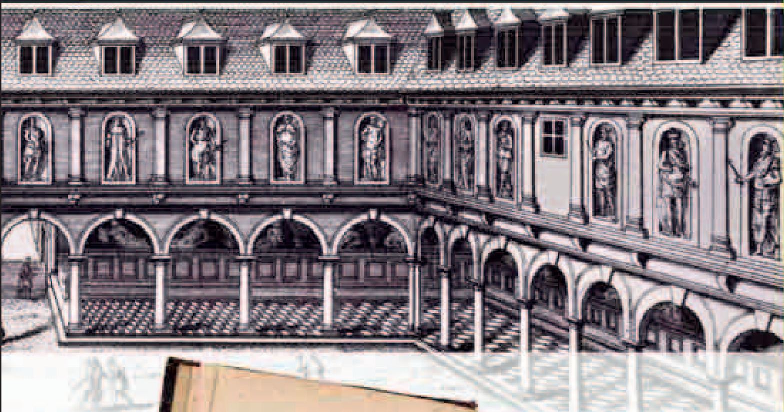
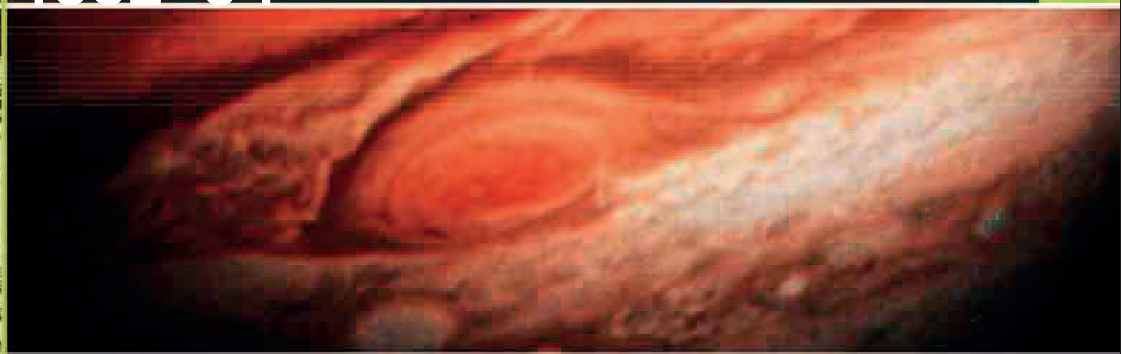
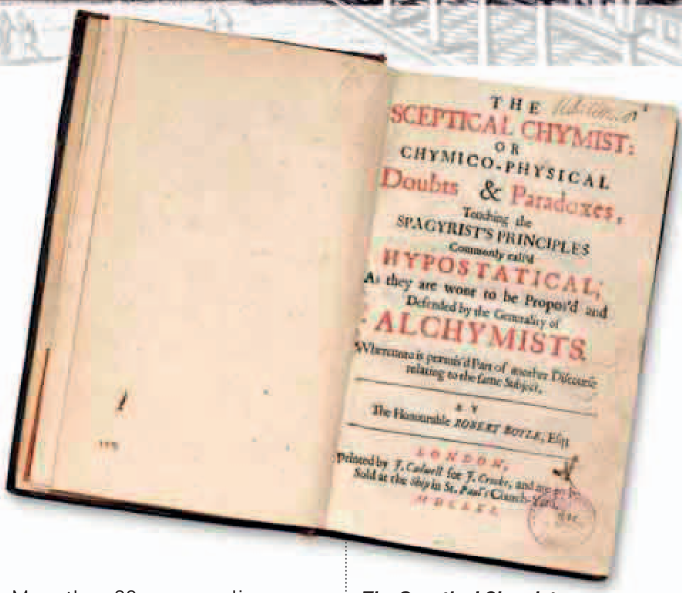




Observed by Robert Hooke in 1664, Jupiter's red spot is a giant storm, large enough to engulf three Earths.



The Sceptical Chymist
Robert Boyle's book is a dialog between fictitious supporters of alchemy and the "voice of reason" extolling a science based on atoms, definable elements, and experiments.

More than 30 years earlier, in his great treatise on blood circulation, English physician William Harvey had suggested that the body contained minute blood vessels that connected arteries with veins—and thereby completed the circuit. In 1661, Italian physician and biologist **Marcello Malpighi** used his microscope and discovered these **blood capillaries**. Malpighi devoted much of his career to the microscopic study of anatomy. He would go on to make important discoveries about the kidneys, embryos, insects, and even plants.

> " I NOW MEAN BY **ELEMENTS,** CERTAIN **PRIMITIVE OR SIMPLE,** OR **PERFECTLY UNMINGLED BODIES…** "

Robert Boyle, English scientist and inventor, from *The Sceptical Chymist*, 1661

TWENTY YEARS after it was proved that air pressure decreases with height, English meteorologist **Richard Towneley** noted that a fixed amount of **trapped air expanded** in volume **at high altitude**. Robert Hooke subsequently confirmed these observations by experimentation. Robert Boyle published what he called "Towneley's hypothesis" in 1662, but later it became known as **Boyle's law**.

In the same year, at a time when an outbreak of the bubonic plague in London was imminent, English shopkeeper **John Graunt** published his analysis of *Bills of Mortality*. Although not a scholar, Graunt used these records to work out **population trends**. Impressed with Graunt's efforts, Charles II ordered the Royal Society to admit him as a fellow. Today, Graunt's **life tables** mark the foundation of the statistical study of populations: **demography**.

In 1663, Scottish astronomer **James Gregory** proposed a design for a **reflecting telescope**. It incorporated mirrors as a way of avoiding the color aberrations that arose when lenses refracted (bent) different wavelengths of light. However, it was Isaac Newton who was able to get the first reflecting telescope made (see 1667–68).

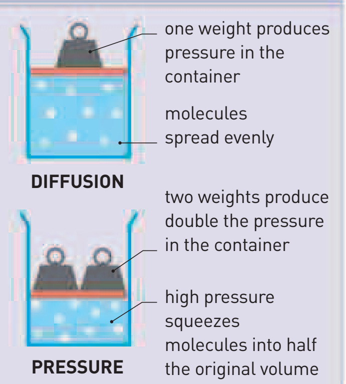
Although astronomers had studied **Jupiter** earlier, they did not record its **Great Red Spot** until the 1660s. This may have been because of inadequate telescopes or because, until then, it was not there at all. The spot, which is a giant storm, probably started only around 1600. **Robert Hooke** observed it in 1664, but Italian astronomer Giovanni Cassini may have seen it as early as 1655.

ROBERT BOYLE (1627–91)

English physicist and inventor, Robert Boyle was a pioneer of chemistry as well. He pursued science through experiment and reasoning, and he was inspired by Galileo's work (see 1611–13). Boyle made an air pump and used it to study the behaviour of gases. One of the first fellows of the Royal Society, he came up with the modern concept of chemical elements.

BOYLE'S LAW

Unlike liquids, gases are compressible. Physicist Robert Boyle formalized a law describing the relationship between the pressure of a gas and its volume. As long as temperature stays the same, the pressure and volume of a gas are inversely proportional. In quantitative terms this means that if pressure is doubled, volume is halved and vice versa.



March 15, 1661 Marcello Malpighi publishes a treatise on the **lung**, describing its **alveoli** (air sacs) and network of **capillaries**

March 1661 Malpighi makes the first unambiguous reference to **blood cells**

1661 Niels Stensen discovers the duct of the largest salivary gland, the **parotid gland**

1661 Robert Boyle publishes ***The Sceptical Chymist***

1662 Lorenzo Bellini describes the **anatomy of the kidneys**

1662 Robert Boyle describes the relationship between volume and pressure of a gas, later called **Boyle's Law**

1662 John Graunt publishes *Natural and Political Observations Made Upon Bills of Mortality*

1663 Nicholas Steno recognizes that the **heart** is made of **muscle**

1663 James Gregory describes the design of a **reflecting telescope**

May 9, 1664 Robert Hooke observes **Jupiter's Great Red Spot**

1664 Thomas Willis publishes *Anatomy of the Brain, with a Description of the Nerves and their Functions*